

Options Price Corporate Filings Correctly: Two Pre-Registered 8-K Tests on the 100 Largest US Stocks

Theo Machado Jacob Crainic

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Abstract

On the day before a company files a Form 8-K, the options on its stock already price how far the stock should move. With rules committed before any data, we tested whether that price is wrong after slow-burning bad news (a bought put should be cheap) and after restructurings (a sold put should be dear). Neither passed. Over 2024 and 2025 the options did not under-price post-filing protection by more than about 3 percent of the stock price, nor over-price restructuring puts by more than about 0.9 percent. On 2022, a year no test had used, the protective put beat ordinary days by 4.3 percent at 21 sessions, but the gain came from stocks rebounding after the news, not from cheap puts.

1 Hypothesis

Our main paper finds that options forecast Polymarket’s stock contracts better than Polymarket does, so we treat the chain as an event’s reference price and ask where it could still be wrong. **H1** (litigation, class actions, regulatory investigations, cyber incidents, goodwill, asset and investment impairments): implied volatility falls once the headline passes, but the legal or accounting damage resolves over weeks, so the chain should over-price the first day and under-price the follow-through, and a **protective put** bought after the filing should be cheap. **H2** (restructuring plans, workforce reductions, facility closures, business-line exits): holders who must keep the stock through the event buy puts, and dealers charge for demand they cannot offset (Gârleanu, Pedersen and Potesman, 2009, *Rev. Financ. Stud.* 22:4259), so a **cash-secured put** sold after the filing should be dear. Each hypothesis predicts both a profit and a sign on the parity ratio R , the realized move divided by the move the chain priced on the session before the filing (H1 above ordinary days, H2 below), so a strategy that is merely long or short volatility cannot pass.

2 Data and method

Filings come from Massive’s 8-K disclosure tags and options from its chains and daily bars, through the starter’s pipeline: the 100 largest US companies, put-call-parity spot, puts 5 percent out of the money with 90 to 180 days to expiry. The 11 tags, the pass rule and a forecast for unseen windows were committed before any event or price was fetched.¹ One event counts per company and filing date; 5 filings tagged in both families are dropped. Every filing is treated as public after the close, so the trade enters at the next session’s close. The baseline is 120 ordinary days per family on the same companies, at least 30 days from their filings. A hypothesis passes if the 97.5 percent bootstrap interval of its edge over ordinary days lies above zero at 2 or more of 21 sessions, 42 sessions and expiry, with R moving the predicted way there; fewer than 2 testable horizons is *insufficient*. In-sample is 2024–2025, out-of-sample January to August 2026, run once after the method was frozen, and 2022 a further fresh year run once.

3 Results

Neither hypothesis passes in any window (Table 1). Over 2024 and 2025, R moved the predicted way at all three registered horizons for H1 and two for H2, but no interval excludes zero, which bounds the mispricing: post-filing protection was not under-priced by more than about 3 percent of the stock price, nor restructuring puts over-priced by more than about 0.9 percent. Figure 1 shows H1’s predicted shape, 0.68 one session after the filing against 0.99 on ordinary days and 1.36 against 1.15 at 42 sessions, inside the events’ interval. Out of sample H2 reversed sign, because 2026 restructurings moved 2.2 to 2.5 times the implied move in their first week, the opposite of H2.

A fresh year. On 2022, run once, H1 beat ordinary days by 4.30 percent at 21 sessions [+0.54, +7.93] on 12 events and met both conditions at that horizon, but not at 42 sessions or expiry, so it does not pass. Splitting the position shows why, and it is not H1’s mechanism: after the bad news the stocks rose 6.1 percent in 21 sessions, with only a quarter falling, against 0.6 percent on ordinary days, while the put alone lost 1.8 percent. In a falling market the

¹Code, frozen tag lists, results and run logs: <https://github.com/theomachado05/polybridge>; notebook research/massive_8k.ipynb, which takes start and end dates and runs a sealed window with RUN_HOLDOUT.

Table 1: Edge over ordinary days on the same stocks, percent of the stock price, with 97.5 percent intervals; bold rows are the registered horizons. H1 in 2026 had 3 events, too few for an interval.

Sessions	H1, protective put		H2, cash-secured put	
	2024–25 (<i>n</i> 28–33)	2022 (<i>n</i> 11–12)	2024–25 (<i>n</i> 23–24)	2026 (<i>n</i> 7)
1	−0.22 [−0.63, +0.19]	+0.25 [−0.37, +0.83]	−0.25 [−0.52, −0.02]	+0.23 [−0.54, +1.03]
2	−0.16 [−1.00, +0.59]	+0.88 [−0.13, +1.88]	−0.17 [−0.51, +0.12]	+0.28 [−0.35, +1.05]
3	+0.08 [−1.09, +1.18]	+0.85 [−0.59, +2.19]	−0.22 [−0.73, +0.21]	−0.01 [−0.77, +0.91]
5	−0.13 [−1.90, +1.75]	+0.82 [−0.67, +2.28]	−0.08 [−0.53, +0.33]	+0.01 [−1.52, +1.44]
10	−0.46 [−2.56, +1.65]	+2.32 [−0.48, +5.34]	+0.05 [−0.62, +0.78]	−0.64 [−2.79, +1.37]
21	+0.07 [−2.70, +2.91]	+4.30 [+0.54, +7.93]	+0.09 [−0.68, +0.89]	−2.15 [−7.59, +2.03]
42	+0.49 [−4.51, +5.77]	+0.79 [−3.51, +5.52]	+0.21 [−0.90, +1.36]	−1.38 [−7.29, +2.60]
63	+3.19 [−2.65, +9.14]	+0.33 [−6.29, +9.13]	+0.65 [−0.32, +1.69]	+3.37 [−4.27, +10.53]
Expiry	+3.33 [−3.01, +10.12]	+1.54 [−6.33, +11.61]	+0.92 [−0.36, +2.34]	(4 events)

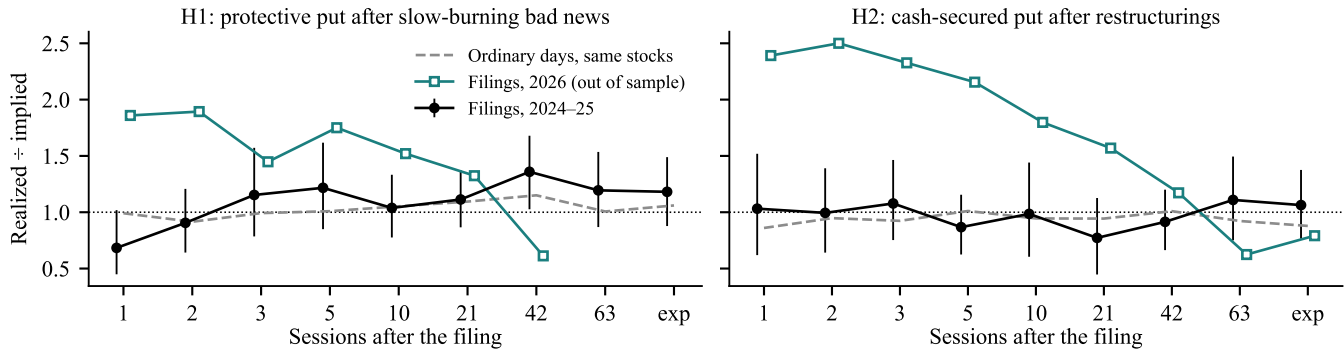


Figure 1: Realized move divided by the move the chain priced on the session before the filing, with 95 percent intervals for 2024–25 filings; 1 means priced correctly.

headline was oversold and the put dear, and the protective put earned through the stock it owns. A filter to the most liquid names, set before the run, left too few events (3 and 1).

4 The rules run as a book

Table 2 runs each registered rule as a trader would have: one trade per filing at the registered cell (3 to 6 month expiry, 5 percent out of the money, bought or sold at the close of the session after the filing, held 21 sessions), each sized at \$1 of stock, net of a 5 percent premium charge each way, and booked on its exit date (Figure 2). Positions overlap, so the Sharpe ratio is per trade, annualized by the window’s trade rate. The ordinary-day column is the same put on the same stocks away from filings, annualized at the same rate and net of the same charge. It is the benchmark, because a put on a large stock earned money in 2024 and 2025 whether or not a filing came first.

In-sample both books made money, with Sharpe ratios of 1.20 and 1.41 against 1.01 and 0.74 for the same puts on ordinary days. The gap is the filing’s contribution, and Table 1 shows it is not distinguishable from zero. Out of sample the H2 book lost 25 percent of one trade’s notional over 7 trades, its worst trade 15.5 percent, as 2026 restructurings fell further than the chain priced. That is the risk a sold put carries and the reason we would size it, if at all, as below.

5 Sensitivity

Every number is from the frozen in-sample run, never chosen from it. At 21 sessions, with the tradeable entry, the H2 edge over ordinary days is positive in all 9 cells of expiry bucket (1, 2, 3 to 6 months) by out-of-the-money distance (3, 5, 10 percent), +0.08 to +1.24 percent, and in all 9 with entry on the session before the filing; H1 is positive in 7 of 9 (−0.67 to +0.68) and 3 of 9 with the earlier entry. The cells share events, so this is a stable sign, not 18 results. Table 1 covers every horizon. On category definition, splitting each family into its tags (exploratory, at most 15 events a tag), only restructuring plans with the sold put clear zero: +2.57 percent at expiry [+0.95, +4.51], a

Table 2: Each rule as a book at 21 sessions, net of costs; returns and drawdowns in percent of one trade's stock notional. Turnover is 12 times a year in every column, because each position lives 21 sessions.

	H1, protective put			H2, cash-secured put		
	2024–25 in-sample	2026 out-of-sample	2022 fresh	2024–25 in-sample	2026 out-of-sample	2022 fresh
Trades (per year)	32 (16)	3 (4.5)	12 (12)	24 (12)	7 (10.5)	12 (12)
Net return per trade	+1.81	+2.68	+3.85	+0.54	−2.82	−0.12
Winning trades	62%	67%	75%	71%	43%	50%
Annualized return	+29.0	+12.1	+46.2	+6.5	−29.6	−1.4
Annualized volatility	24.1	17.1	19.8	4.6	21.8	10.6
Sharpe ratio, net	1.20	0.71	2.33	1.41	−1.36	−0.13
Sharpe, same put on ordinary days	1.01	0.34	−0.19	0.74	−0.25	−0.88
Maximum drawdown	−9.6	−5.5	−8.2	−2.8	−25.3	−12.1
Worst trade	−9.3	−5.5	−5.4	−2.0	−15.5	−8.1
Positions open on average	1.3	0.4	1.0	1.0	0.9	1.0

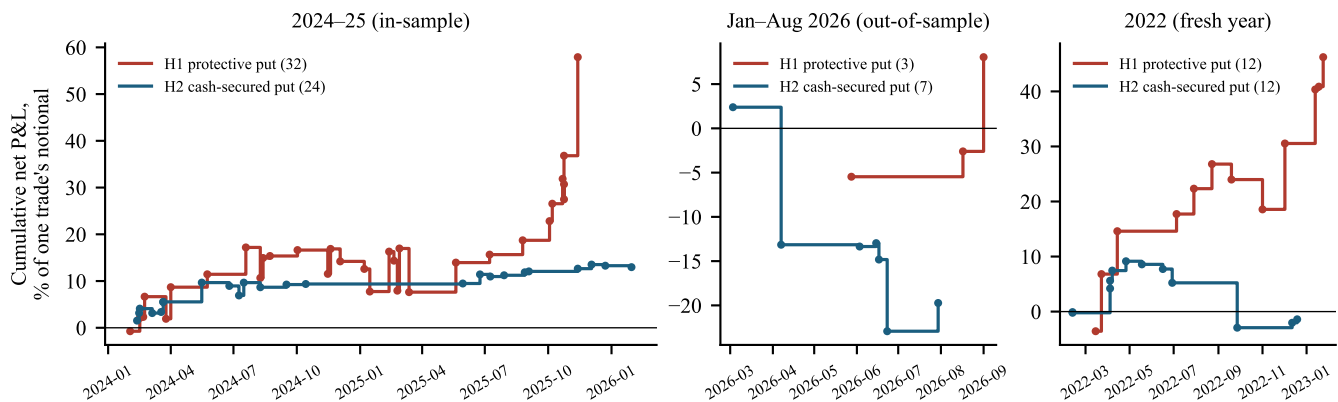


Figure 2: Cumulative net profit of each rule, one \$1 trade per filing, booked at exit; trade counts in the legend.

lead for a new registered test. **Variants tested:** two registered rules decide the verdict; the notebook's grid shows 810 cells per tag (3 buckets, 2 entries, 3 distances, 5 strategies, 9 horizons) and the tag atlas 11 tags, all reported and none selected.

6 Liquidity, costs and capital

Entry respects the filing lag: every filing is treated as public after the close, so the order goes in at the next session's close, which is late for filings made before the open. A 5 percent premium charge each way costs 28 to 29 basis points of the stock price; quoted half-spreads, where Massive has quotes, are 26 to 29, so the charge is close to crossing the spread. Net of it the in-sample edge over ordinary days is about +11 basis points for both families. The median put traded 34 (H1) and 49 (H2) contracts on the entry day, so at 10 percent of that volume a trader fills 3 to 5 contracts, about \$0.1 million of stock, per filing. With about one filing per family a month and 1 to 1.3 positions open, the strategy deploys about \$0.15 million: the protective put ties up the stock plus a premium near 2 percent of it, and the cash-secured put the strike in cash. At in-sample returns that is a few thousand dollars a year before any edge over ordinary days, so even a real edge would be a cost saving for a hedger, not a fund.

7 Risk management

Were either rule traded, these limits are fixed before the first order. **Size:** equal stock notional per filing, no more than 10 percent of the put's entry-day volume and no more than a fifth of capital on one name. **Tail:** the protective put's loss is capped at the premium plus 5 percent of the stock; the cash-secured put is not, so the H2 book holds the strike in cash (no margin), caps the sum of open strikes at the capital, and keeps at most two names from one sector. **Overlap:** a filing within 10 sessions of an earnings date is skipped, because the earnings move would swamp the

filing's. **Stop:** a rule stops trading when its live edge over ordinary days, re-estimated each quarter on new filings, has an upper 95 percent bound below zero, or when its drawdown exceeds the in-sample maximum by half; out of sample this would have stopped H2 after its second trade, a 15.5 percent loss on Nike against a -2.8 percent in-sample maximum, and avoided a further -6.6 percent over its last five. **Data:** an order is refused when the option's last trade is older than a day or the put-call parity spot differs from the stock's close by more than 1 percent.

8 What would break it, and what we predicted

Samples are small, the company list is today's, spot is inferred from the chain and marks are daily last trades. Before running any unseen window we committed a forecast: it called no pass and the H2 count out of sample, and missed the H1 count (3 against about 11), the interval widths and all three signs it could score. **For the judges' sealed window it predicts** about 4 H1 and 3 H2 events in three months, so *insufficient* for both, and a null for both in six months or more; a pass would contradict it. The notebook prints its verdict beside this forecast.

How to trade it. We would not trade either rule as a strategy. A portfolio manager who wants protection after these filings can buy it at the next close without overpaying by more than the bounds in Section 3, and a holder writing puts after a restructuring is not paid extra for it. Next tests, each committed before its data: restructuring plans with the sold put, H2 entry after the first-week move, and post-filing rebounds in falling markets.